

Jiulong Lin

Senior Frontend-Focused Full-Stack Engineer

Cupertino, CA · (347) 865-6717 · longgangsima@gmail.com

SUMMARY

Senior Frontend-Focused Full-Stack Engineer with 5 years building real-time, data-intensive web applications at Walmart and Sam's Club. Strong in React, TypeScript, GraphQL, and backend API design, with a focus on end-to-end system design, data consistency, and real-time synchronization across frontend and backend boundaries.

Stack: **React 18, TypeScript, Next.js** · Redux Toolkit, TanStack Query, Apollo Client · **GraphQL, REST APIs** · GraphQL, REST APIs · Node.js, **Python**, Django, PostgreSQL · HTML5, CSS3/SCSS, Tailwind · Frontend Architecture, Real-time State Management · Jest, RTL, Playwright · Jenkins, GitHub Actions

EXPERIENCE

Walmart — Compass | Frontend / Full-Stack Engineer (Contract)

Cupertino, CA | Mar 2025 – Present

- Designed and built compass-validator end-to-end (Node.js/TypeScript/Playwright) — automated validation across 9,000+ stores in CI, replacing manual Chrome-based sampling. Architected a producer-consumer worker pool for skew-tolerant parallelism, plus a RetryManager that distinguishes transient Playwright flakes from deterministic map defects — failures whose root cause shifts across retries are flagged for investigation rather than being masked by retry logic.
- Published as an internal **NPM package adopted by 3 teams**; introduced a **ConfigProvider abstraction** (selectors, thresholds, and rule sets) that let the Glass team adopt with their own map variant — different DOM structure, different validators — **without forking the core engine**.
- Integrated compass-validator into the Kafka-orchestrated Map Refresh Pipeline as a pre-persist validation gate, surfacing per-rule diagnostics to downstream systems and dashboard UI; designed an end-to-end failure traceability flow capturing per-rule failure reasons, extending the pipeline's payload, and surfacing them in the dashboard's map detail view.

Sam's Club | Frontend / Full-Stack Engineer (Contract)

Edison, NJ

Sam's Ad Center

Sep 2024 – Jan 2025

- Engineered multi-step campaign workflows with complex field dependencies and cross-field state synchronization while restoring automation pass rate from 40% → 90% by diagnosing systemic failures and building a centralized config system that decoupled Walmart and Sam's environments, eliminating 90% of environment-specific issues.
- Designed a test reliability strategy that isolates environment-specific failures from deterministic defects, reducing flaky test noise and improving signal quality for CI pipelines.

AdPal (Media Planning Platform)

Mar 2023 – Sep 2024

- Owned AdPal frontend as primary frontend engineer for 18 months — a real-time collaborative media planning platform spanning calendar booking, event management, and campaign workflows, replacing an error-prone Excel process responsible for duplicate bookings and revenue leakage.
- Reframed an apparent data-fetching problem as a distributed state consistency problem — concurrent planners could operate on stale capacity within polling windows. Pushed back on REST-optimization quick fixes; led an incremental, reversible REST→GraphQL subscription migration with version-aware writes, establishing a consistent real-time update model across concurrent planners — cut API traffic ~80% and eliminated stale-data duplicate bookings.
- Designed a **two-query architecture** (Apollo split() + InMemoryCache typePolicies) eliminating **~22,000 fields** of per-poll over-fetching, with **TanStack Query as the single source of truth** and Apollo scoped strictly as a subscription transport sidecar — keeping cache ownership separate from transport.
- Restored **WebSocket subscription reliability ~60% → ~99%** by diagnosing four root causes (heartbeat drops, async token refresh, reconnect catch-up, version-aware cache dedup); paired with **optimistic concurrency control** (version IDs) and a **sync-mode state machine** falling back to polling under transport degradation.
- Designed and implemented **Python backend APIs using Django + PostgreSQL**, supporting REST and GraphQL workflows for blackout management and event detail workflows, including a custom audit-log system capturing field-level diffs and actor attribution — ensuring consistency between backend state transitions and real-time frontend updates.

Opus (CMS) | Sam's Club

Mar 2022 – Mar 2023

- Built **Design Pages** and **Design Modules** management interfaces for Opus, the CMS powering SamsClub.com, letting content teams configure reusable module templates with targeting rules across multiple page types.
- Delivered the editorial toolset — **File Manager** with hierarchical folders and per-file **role-based access control**, drag-and-drop layout composition, **WYSIWYG preview** integrated into the draft→staging→publish workflow, and version history.

Walmart — Global Experience | Frontend Engineer (Contract)

Edison, NJ | Mar 2021 – Mar 2022

- Delivered **search and homepage micro-frontend modules for Walmart.ca**, serving **millions of Canadian shoppers** within an orchestration + event bus architecture consumed by iOS/Android SDK clients via WebView; achieved **WCAG 2.1 compliance**; configured Dynatrace alerting + xMatters on-call; CI/CD test coverage spanning Jest, RTL, CodeceptJS, Playwright.

EDUCATION

M.S. Engineering Management, Trine University **B.S. Applied Mathematics & Statistics** (Minor: CS), Stony Brook University